

Mathematical Concepts

for (MATH101, 102, 311, 312) Calculus and Analysis

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This document collects mathematical concepts for (MATH101) Calculus I, (MATH102) Calculus II, (MATH311) Analysis I, and (MATH312) Analysis II. General mathematical notation and writing follow *(MATH000) Mathematical Language*.

Contents

Notation and Conventions	2
Part 1 (MATH101) Calculus I	5
Part 2 (MATH102) Calculus II	6
Part 3 (MATH311) Analysis I	7
Part 4 (MATH312) Analysis II	8

Notation and Conventions

General mathematical notation and writing follow (*MATH000*) *Mathematical Language*. The conventions below record only notation specific to calculus and analysis used in this book. When several equivalent notations are listed, the first form is used throughout.

Single-Variable Calculus

The differential d is upright, and ε is the default small positive quantity in analysis.

Notation	Meaning
$\lim_{x \rightarrow a} f(x)$	limit as x approaches a
$\lim_{x \rightarrow a^-} f(x)$	left-hand limit as x approaches a
$\lim_{x \rightarrow a^+} f(x)$	right-hand limit as x approaches a
$\min A$	minimum of A
$\max A$	maximum of A
$\inf A$	infimum of A
$\sup A$	supremum of A
$f'(a)$	first derivative of f at a
$f^{(n)}(a)$	n th derivative of f at a
$\frac{dy}{dx}$	derivative of y with respect to x
$\int_a^b f(x) \, dx$	definite integral over $[a, b]$

Elementary Functions

Notation	Meaning
e^x , $\exp x$	exponential function
$\ln x$, $\log_e x$	logarithm to base e
$\lg x$, $\log_{10} x$	logarithm to base 10
$\text{lb } x$, $\log_2 x$	logarithm to base 2
$\log_a x$	logarithm to base a
$\sin x$	sine
$\cos x$	cosine
$\tan x$	tangent
$\csc x$	cosecant
$\sec x$	secant
$\cot x$	cotangent
$\arcsin x$, $\sin^{-1} x$	principal inverse sine
$\arccos x$, $\cos^{-1} x$	principal inverse cosine
$\arctan x$, $\tan^{-1} x$	principal inverse tangent
$\text{arccsc } x$, $\csc^{-1} x$	principal inverse cosecant
$\text{arcsec } x$, $\sec^{-1} x$	principal inverse secant
$\text{arccot } x$, $\cot^{-1} x$	principal inverse cotangent
$\sinh x$	hyperbolic sine
$\cosh x$	hyperbolic cosine
$\tanh x$	hyperbolic tangent

Notation	Meaning
$\operatorname{csch} x$	hyperbolic cosecant
$\operatorname{sech} x$	hyperbolic secant
$\operatorname{coth} x$	hyperbolic cotangent
$\operatorname{arcsinh} x, \operatorname{arsinh} x,$ $\sinh^{-1} x$	principal inverse hyperbolic sine
$\operatorname{arccosh} x, \operatorname{arcosh} x,$ $\cosh^{-1} x$	principal inverse hyperbolic cosine
$\operatorname{artanh} x, \operatorname{atanh} x,$ $\tanh^{-1} x$	principal inverse hyperbolic tangent
$\operatorname{arcsch} x, \operatorname{arsch} x,$ $\operatorname{csch}^{-1} x$	principal inverse hyperbolic cosecant
$\operatorname{arcsech} x, \operatorname{arsech} x,$ $\operatorname{sech}^{-1} x$	principal inverse hyperbolic secant
$\operatorname{arcoth} x, \operatorname{coth}^{-1} x,$ $\operatorname{coth}^{-1} x$	principal inverse hyperbolic cotangent
$\Gamma(z)$	Gamma function

Series and Convergence

Notation	Meaning
$\sum_{k=1}^n a_k$	finite sum
$\sum_{k=1}^{\infty} a_k$	infinite series
$\prod_{k=1}^n a_k$	finite product
$x_n \rightarrow x$	convergence of a sequence
$a_n \nearrow L$	monotone increasing convergence to L
$a_n \searrow L$	monotone decreasing convergence to L
$\limsup a_n$	limit superior
$\liminf a_n$	limit inferior
$f_n \rightarrow f, f_n \xrightarrow{\text{ptw}} f$	pointwise convergence
$f_n \rightrightarrows f, f_n \xrightarrow{\text{unif}} f$	uniform convergence
$f_n \xrightarrow{\text{a.e.}} f$	convergence almost everywhere
$C^k(I)$	k times continuously differentiable functions on I
$C^\infty(I)$	smooth functions on I
$\ f\ _\infty$	supremum norm
$B_r(x)$	open ball of radius r centered at x
$\overline{B}_r(x)$	closed ball of radius r centered at x

Multivariable Calculus

Notation	Meaning
$\frac{\partial f}{\partial x_i}, \partial_i f$	partial derivative with respect to x_i
$D\mathbf{F}(\mathbf{a})$	total derivative; Jacobian matrix in coordinates
$\operatorname{Jac}(\mathbf{F})$	Jacobian determinant

Notation	Meaning
$\text{Hess } f$	Hessian matrix of f
∇f	gradient of f
$\nabla \cdot \mathbf{F}$	divergence of $\mathbf{F}$
$\nabla \times \mathbf{F}$	curl of $\mathbf{F}$
Δf	Laplacian of f
$\iint_D f \, dA$	double integral over D
$\iiint_\Omega f \, dV$	triple integral over Ω
$\mathbf{r}(t)$	parametrized curve
ds	element of arc length
$\int_C \mathbf{F} \cdot d\mathbf{r}$	line integral along C
$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$	flux integral through S

Part 1

(MATH101) Calculus I

Part 2

(MATH102) Calculus II

Part 3

(MATH311) Analysis I

Part 4

(MATH312) Analysis II
